

Analytical Question 1: Using optimization through clustering methods and an objective function, with province population density and geographic distance, where are the next recommended coordinates for new sensor deployment?

Xi, J., Zhang, B., & Yang, Y. (2024). Optimizing air quality monitoring spatial layout by maximizing the coverage of the population in Beijing–Tianjin–Hebei and surrounding areas. *Science of the Total Environment*, 955, 177029. <https://doi.org/10.1016/j.scitotenv.2024.177029>

Analytical Question 2: Using time series analysis, what are the defining trends and features of PM2.5 concentrations during historical peak years, and how can these patterns call for preemptive health advisories and prioritized monitoring?

Lu, X., Liao, J., Huang, H., Li, J., Huang, H., & Zhu, Y. (2025). Multi-scale dynamic graph neural network for PM2.5 concentration prediction in regional station cluster. *PLOS One*, 20(12), e0338392. <https://doi.org/10.1371/journal.pone.0338392>

Results—OpenURL Connection—EBSCO. (2026). Ebsco.Com. <https://openurl.ebsco.com/contentitem/gcd:196706195?sid=ebsco:plink:crawler-gcd&id=ebsco:gcd:196706195&crl=c&jrnl=2158107X>

Thangavel, P., Park, D., & Lee, Y.-C. (2022). Recent Insights into Particulate Matter (PM2.5)-Mediated Toxicity in Humans: An Overview. *International Journal of Environmental Research and Public Health*, 19(12). <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9223652/>

Wang, Y., & Du, R. (2026). Same-Hour Estimation and Multi-Horizon Forecasting of PM2.5 in Beijing: Model Comparison and Feature-Group Contributions (2607.07279v3). arXiv. <https://arxiv.org/abs/2607.07279v3>

World Health Organization. (2021, September). C40 Knowledge Community. [Www.C40knowledgehub.Org. https://www.c40knowledgehub.org/s/article/WHO-Air-Quality-Guidelines?language=en_US](https://www.c40knowledgehub.org/s/article/WHO-Air-Quality-Guidelines?language=en_US)

Zhong, J., Zhang, X., Gui, K., Wang, Y., Che, H., Shen, X., Zhang, L., Zhang, Y., Sun, J., & Zhang, W. (2021). Robust prediction of hourly PM2.5 from meteorological data using LightGBM. In *National Science Review* (Vol. 8, Issue 10, p. nwaa307). PubMed Central. <https://doi.org/10.1093/nsr/nwaa307>

Analytical Question 3: Using predictive modeling, which existing sensor location should be prioritized for increased maintenance and monitoring frequency using a predicted risk score calculated from downtime durations and consistent outlier behavior?

Budiman, D. A., Sutikno, T., & Fadlil, A. (2026). Efficient Outlier Detection in Energy Analytics Using Isolation Forest and One Class SVM: A Comparative Study for Smart Grid Applications. *Jurnal Accounting Information System (AIMS)*, 9(1), 87–98. <https://doi.org/10.32627/aims.v9i1.1913>

DENR: Establishment Of An Integrated Air Quality Network Center That Monitors And Serves As Repository Of Real-Time Industrial Emission <https://air.emb.gov.ph/wp-content/uploads/2021/11/EMB-MC-2021-14-Signed.pdf>

Hoang, N., & Pham. (2025). Exploring ARIMA and TimeGAN combined with ECOD for Anomaly Detection in Time Series: A Focus on RealKnownCause in the NAB Dataset.

<https://helda.helsinki.fi/server/api/core/bitstreams/51b1979a-49a3-4fd2-8ad3-d448f2246bfc/content>

Mojžišová, I. (2025, November 5). Application of the Fourier Transform in data preparation for AI: Beyond the time domain. ERNI.

<https://www.betterask.erni/application-of-the-fourier-transform-in-data-preparation-for-ai-beyond-the-time-domain/>

Wong, L., Liu, D., Berti-Equille, L., Alnegheimish, S., & Veeramachaneni, K. (n.d.). AER: Auto-Encoder with Regression for Time Series Anomaly Detection. Retrieved September 25, 2026, from https://laureberti.github.io/website/pub/2022_IEEE_BigData.pdf

Column Name	Data Type	Data Type Classification	Unit or Format	Number of Nulls	Number of Non-nulls	Technical Description	Business Meaning
station	String	Categorical	Alphanumeric	0	1027171	Official designation of the air quality monitoring station.	Anchors sensor readings to a physical hardware asset. Used to track sensor uptime, isolate localized pollution spikes, and group longitudinal station trends.
lat	Float	Numerical	Decimal Degrees	0	1027171	North-south geographic coordinate of the monitoring station.	Spatial Coordinate
datetime_utc	DateTime	TimeStamp	YYYY-MM-DD HH:MM:SS (UTC)	0	1027171	Exact date and time the sensor reading was recorded, standardized to Coordinated Universal Time (UTC).	Enables time-series forecasting, seasonality analysis (e.g., dry vs. wet season), and diurnal cycle tracking (e.g., converting to local time to isolate rush-hour

							traffic impact).
pm25 in $\mu\text{g}/\text{m}^3$	Float	Numerical	Micrograms per cubic meter	0	1027171	Atmospheric concentration of fine inhalable particulate matter with aerodynamic diameters of 2.5 micrometers or smaller.	Primary indicator of air quality and respiratory health risk. Drives Air Quality Index (AQI) categorization, regulatory threshold alerts (e.g., WHO or DENR guidelines), and public health exposure modeling.
lon	Float	Numerical	Decimal Degrees	0	1027171	East-west geographic coordinate of the monitoring station.	Spatial Coordinate
city	String	Categorical	Standardized Text	0	1027171	Official name of the city or municipality where the monitoring station is situated.	Primary grouping level for municipal policy-making, local government unit (LGU) reporting, and city-to-city environmental benchmarking.
province	String	Categorical	Standardized Text	0	1027171	Official name of the province or regional administrative division encompassing the city.	Supports macro-level environmental planning, inter-provincial pollution transport analysis, and regional budget/resource allocation.

population	Integer	Numerical	Whole Number	0	1027171	Total resident population of the host city or municipality based on the latest official census.	Quantifies the human footprint at risk. Used to compute population-weighted PM2.5 exposure to prioritize interventions where high pollution overlaps with large populations.
area_in_sq.km	Float	Numerical	Square Kilometers	0	1027171	Total official land area of the city or municipality in square kilometers.	Normalizes municipal metrics across jurisdictions of varying physical sizes and helps evaluate monitoring coverage adequacy (e.g., stations per km ²)
density_persons/sqkm	Integer	Numerical	Persons per kilometer square	0	1027171	Ratio of total resident population to total municipal land area.	Serves as a proxy for human activity intensity (vehicular congestion, residential emissions, commercial density). Critical feature for modeling how urban crowding correlates with baseline particulate pollution.

Group Number 2
Group Members Cuyo, Labilles, Maranion, Ong, Toliba, Tumaneng
Course Prescriptive Analytics_DSS142P_BM17

Data Dictionary

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4	pm25 in µg/m³	Float	Numerical	Micrograms per cubic meter	0	1027171	Atmospheric concentration of fine inhalable particulate matter with aerodynamic diameters of 2.5 micrometers or smaller.	Primary indicator of air quality and respiratory health risk. Drives Air Quality Index (AQI) categorization, regulatory threshold alerts (e.g., WHO or DENR guidelines), and public health exposure modeling.
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